

HECATE SERVICES · WORKING DESIGN

DroneX

Federated counter-UAS airspace awareness on the Macula capability mesh. Turning a fleet of cheap, heterogeneous sensors into one coherent airspace picture, with no cloud and no single point of failure.

hecate-services/hecate-dronex

Status: Draft / Concept

2026-06-09

Scope: detect · classify · track · alert. Defensive
situational awareness only. No interference with
aircraft.

DroneX: Drone Detection on the Macula Capability Mesh

Status: Draft / Concept · Repo: `hecate-services/hecate-dronex` · Date: 2026-06-09

Scope statement. This document covers **defensive airspace situational awareness**: detect, classify, track, alert. It does not cover, propose, or enable interference with aircraft (jamming, spoofing, takeover, kinetic response). Those belong to a separate legal and ethical regime and are out of scope.

1. Executive summary

Drone detection is fundamentally a **distributed sensor-fusion** problem. No single sensor type is reliable on its own, so a usable system needs many cheap, heterogeneous sensors fused into one coherent airspace picture. That is precisely the shape of a **capability mesh**.

Macula is a good fit, with one honest framing caveat: **Macula is the nervous system, not the senses**. It provides federated discovery, routing, pub/sub and streaming across a fleet of sensors. The actual detection (RF demodulation, Remote-ID decode, acoustic ML, vision inference) is application logic built on top. Macula does not detect drones; it lets a swarm of detectors act as one system without a cloud and without a single point of failure.

This aligns strongly with the strategic positioning of the wider project: European, federated, sovereign-by-construction, public-sector and critical-infrastructure oriented. Counter-UAS airspace awareness is an active European procurement and grant area.

2. What drone detection actually needs

Five detection modalities, all distributed by nature:

Modality	Sensor	What it emits
RF	SDR / protocol sniffer	control-link and telemetry signatures (e.g. DJI OcuSync, Lightbridge)
Remote ID	cheap ESP32 / WiFi+BT receiver	legally mandated drone broadcast ID (EU direct remote ID, FAA Remote ID)
Acoustic	microphone array + classifier	rotor signature, bearing
Radar	micro-Doppler module	range, velocity, track
EO/IR	camera + computer-vision model	visual confirmation, track, payload identification

No single sensor is reliable alone. RF goes quiet on autonomous flight; acoustic struggles in wind and noise; cameras need line of sight and good light; radar is costly. **You need many cheap nodes fused into a track.** That is the entire case for a federated mesh.

The Remote-ID wedge

The cleanest entry point is **Remote ID**. Drones are legally required to broadcast an identifier over WiFi/Bluetooth (EU direct remote identification; FAA Remote ID). A roughly five-euro ESP32 can receive it. A mesh of cheap Remote-ID receivers covering a perimeter becomes a distributed airspace-awareness network at near-zero per-node cost. Macula is the glue that makes two hundred of them behave as one system.

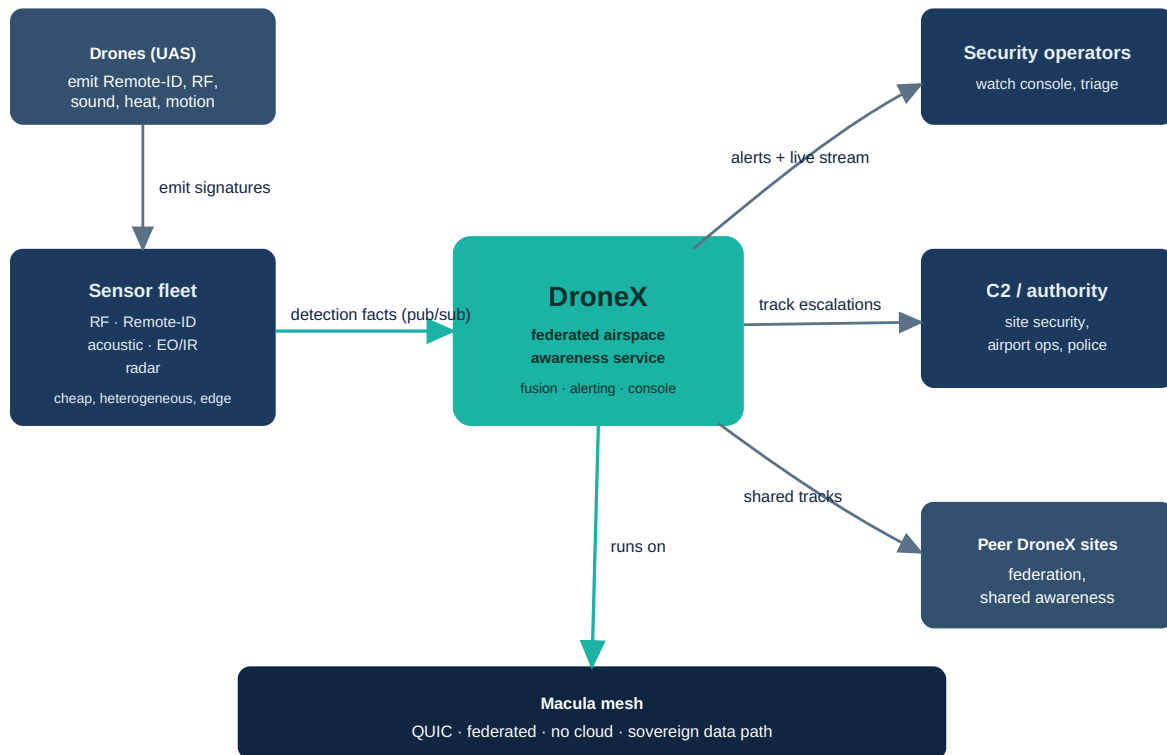
This gives a concrete, demonstrable, low-bill-of-materials first product with a clear public-interest framing and an obvious funding story.

3. Architecture context (C4 Level 1)

DroneX sits between a fleet of edge sensors and the people and systems that act on airspace events. It runs **on** the Macula mesh and federates with peer DroneX sites.

DroneX · System Context

Federated counter-UAS airspace awareness · detect · classify · track · alert



C4 Level 1 (Context). Teal = mesh-native flow. Scope: detect / classify / track / alert only.

DroneX system context

External actors and systems

- **Sensor nodes** (RF, Remote-ID, acoustic, EO/IR, radar) feed detections in.
- **Drones** emit Remote-ID broadcasts and RF/acoustic/optical signatures, the phenomena being observed, not parties to the system.
- **Security operators** receive alerts and view the live console.
- **C2 / external authority** (site security, airport ops, police) receive escalations.
- **Peer DroneX sites** share confirmed-track situational awareness across the federation.
- **Macula mesh** is the substrate everything rides on.

4. Primitive mapping

Macula's four primitives map almost one-to-one onto a counter-UAS pipeline.

Macula primitive	DroneX use
Pub/Sub	Detection facts fan out. Each sensor publishes <code>airspace.contact_observed</code> (bearing, confidence, timestamp, sensor-id). Fusion, alerting and logging subscribe independently.
RPC	On-demand interrogation. "Give me your last 30s of spectrum / IQ capture / a still frame." Fusion calls a sensor only when it needs corroboration, instead of every sensor firehosing everything.
Content Streaming	Live EO/IR video, audio, or RF waterfall to an operator console when a contact escalates.
Content Sharing	Archived clips, captured IQ blobs, evidence packages, content-addressed, suitable for chain of custody.
Capability advertisement	Dynamic, heterogeneous fleet. A node announces <code>rf_sensor / remote_id_sensor / eo_camera</code> . Fusion discovers what is online without static config.

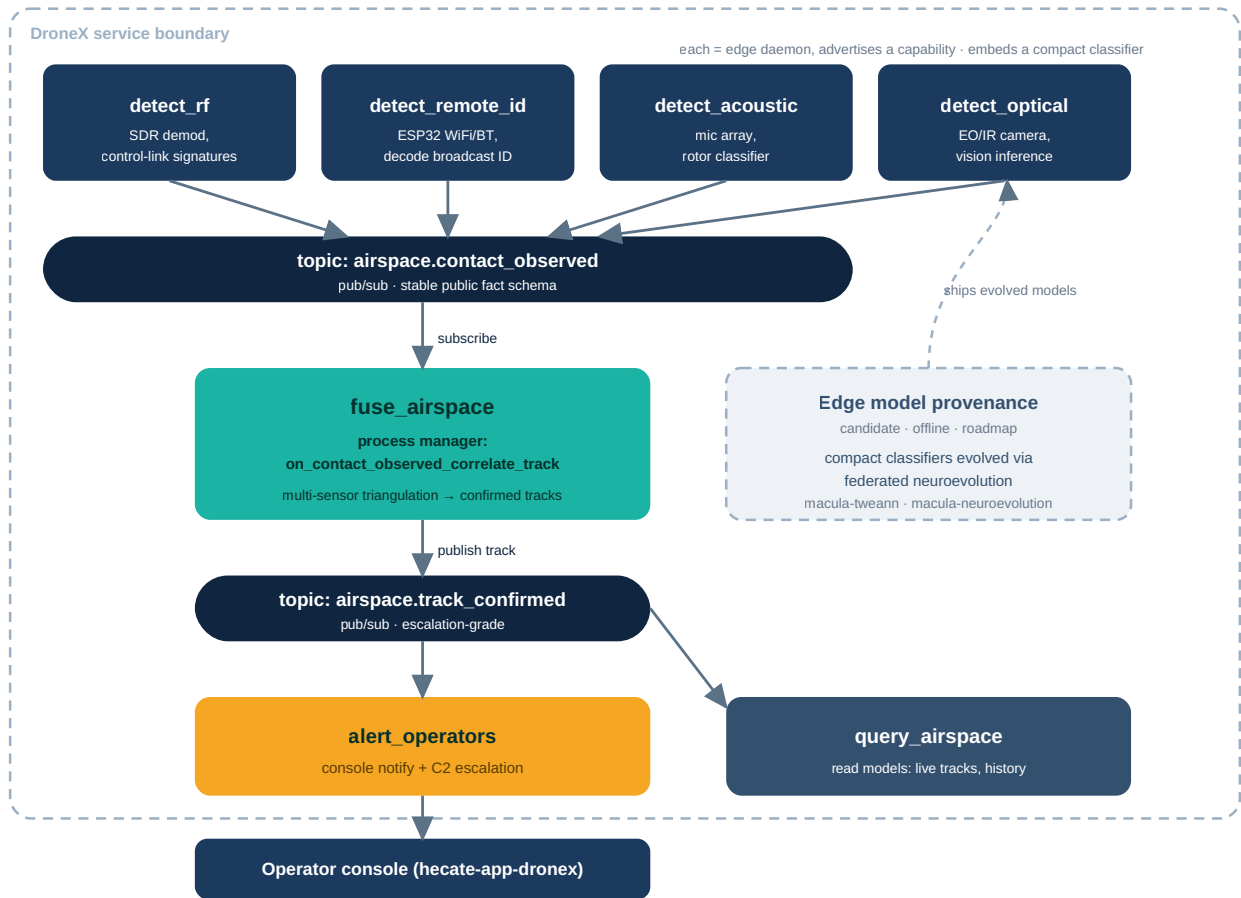
The default carries **tiny detection facts**; raw media moves only on escalation and only by explicit request. This keeps the hot loop cheap.

5. Container model (C4 Level 2) and DDD shape

This is a textbook fit for the Hecate process-centric architecture. Each sensor type is a **Division** (bounded context) owning its full detection lifecycle; a fusion division correlates facts into tracks; an alerting division notifies operators.

DroneX · Container Model

Divisions (bounded contexts) over the mesh · domain events stay internal, facts cross the mesh



C4 Level 2 (Container). Sensor adapters at the edge; fusion + alerting on the edge cluster. Hot loop stays local; mesh shares confirmed tracks.

DroneX container model

Divisions

```

detect_rf      (Division)  ┌
detect_remote_id (Division) ┤ each owns its detection lifecycle
detect_acoustic  (Division) ┤ events: contact_detected_v1,
detect_optical   (Division) ┤ contact_classified_v1,
                           └─┘ contact_lost_v1
                           |
                           | (each publishes an integration FACT to the mesh)
                           ▼
topic: airspace.contact_observed (stable public schema)
                           |
                           ▼
fuse_airspace (Division)
  on_contact_observed_correlate_track (process manager)
  → publishes airspace.track_confirmed
                           |
                           ▼
alert_operators (Division)
  subscribes track_confirmed → console / C2 notification
  
```

Domain events vs integration facts

Detection events stay **internal** (ReckonDB domain events). What crosses the mesh is an explicit **integration fact** with a stable public schema. A sensor swapping its internal model must not break the fusion contract. Counter-UAS is a clean illustration of why that boundary exists.

Naming

Business verbs only, no CRUD: `contact_detected`, `contact_classified`, `contact_lost`, `track_confirmed`, `track_lost`. CMD apps are named for the process (`detect_remote_id`, `fuse_airspace`), not for data management. Query side is `query_airspace`.

Cross-division integration

Fusion reacts to sensor facts through a **process manager** (`on_contact_observed_correlate_track`), never by direct dispatch. The `on_*` directory makes the integration point scream at the filesystem level.

6. Why Macula specifically (not Kafka + a cloud)

1. **Federated, no cloud, survives contested comms.** Counter-UAS at a border, prison, port or airport often runs degraded or air-gapped. A centralized broker is a single kill point. The mesh keeps fusing locally when the uplink is cut.
2. **Edge-native.** Hecate daemons already run on edge devices. A sensor node is just a daemon advertising a capability.
3. **Sovereign data path.** No Microsoft / AWS / Azure in the chain. For European defence and critical-infrastructure procurement this is frequently a hard requirement, not a preference.
4. **Edge AI that fits the problem (and is not an LLM).** Detection is compact, discriminative, per-modality classification: acoustic signatures, RF modulation, micro-Doppler, vision. Those classifiers run **inside each** `detect_*` **slice** (the model belongs to the slice that owns the modality), doing inference at the edge with raw data staying local. This is not a language model, and there is no separate horizontal “AI service.”
5. **Federated training is a natural fit (candidate: neuroevolution).** To produce and improve those compact classifiers, federated neuroevolution (`macula-tweann` / `macula-neuroevolution`, the Faber lineage) is the leading candidate: TWEANN evolves *small* network topologies, which is exactly what cheap edge nodes need, and it can run across the fleet without centralizing raw RF/video. Honest caveat: this is a candidate, not a settled choice. Conventional small supervised models are the baseline; pick empirically, per modality.

7. Honest gaps and mitigations

Gap	Reality	Mitigation
Latency	Cross-station wide-area routing has been a real tuning surface (advertise-settle timing, observer mailbox bottlenecks). Counter-UAS wants sub-second to seconds.	Run fusion at the edge cluster . Use the mesh for cross-site situational sharing, not for the hot detection loop.
High-bandwidth streaming at scale	Raw IQ and 4K EO over the streaming primitive is unproven at fleet scale.	Sensors do local DSP/inference and publish detections (tiny). Stream raw media only on escalation.
Macula is coordination, not algorithms	The RF demod, acoustic classifier and CV model must still be built or bought.	Treat detection algorithms as first-class engineering, not “just integration.”
Edge-AI method unproven	Neuroevolution is an attractive candidate (compact, federated-friendly, on-brand) but not validated for these modalities.	Baseline with conventional small supervised models; benchmark neuroevolution per modality; choose on evidence, not aesthetics.
Dual-use authorization	Detection is defensive and legitimate; anything past awareness is a different regime.	Hard scope: detect / classify / track / alert. No jamming, spoofing, takeover, or kinetic response.

8. Strategic note

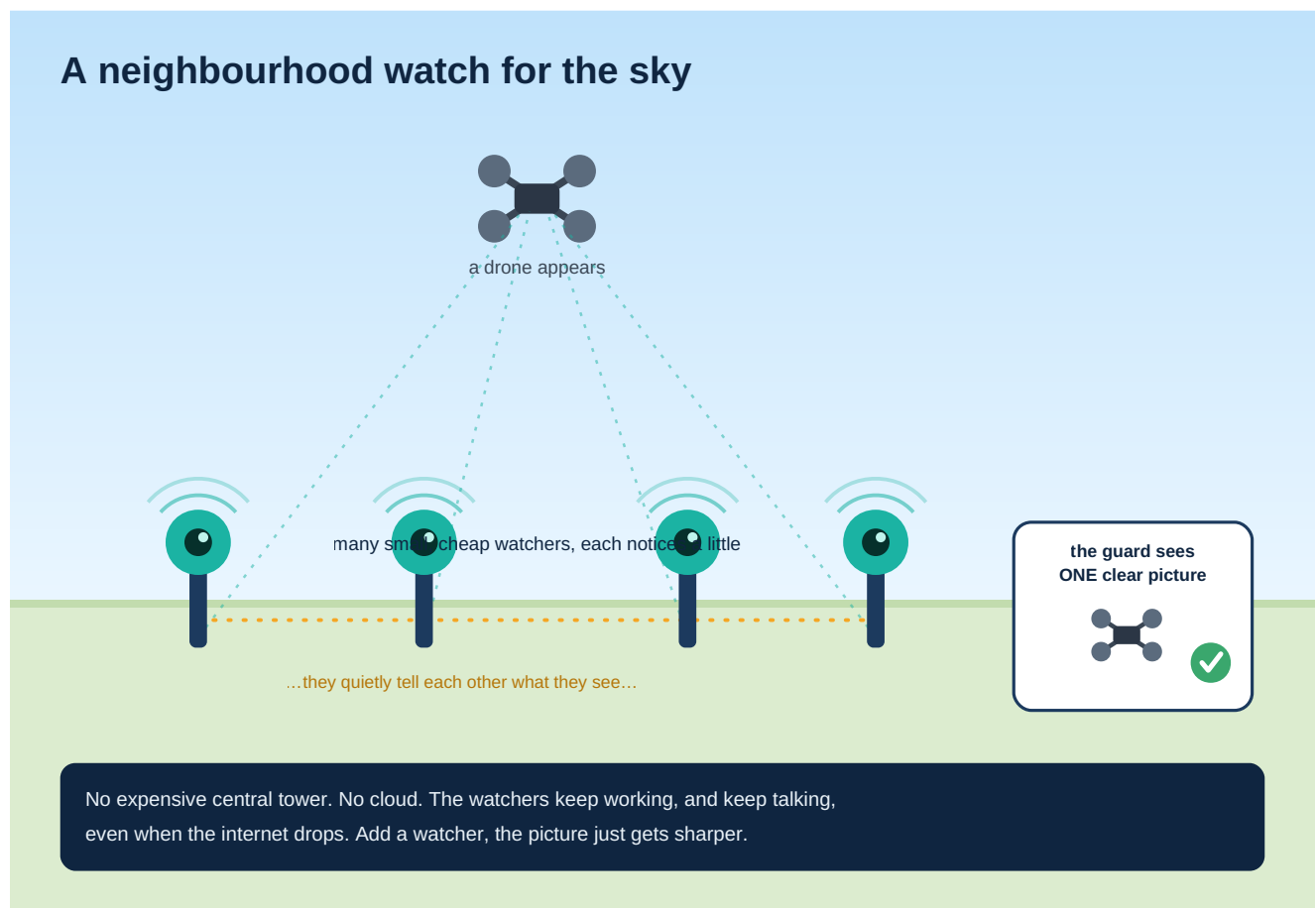
This aligns hard with the project’s positioning: European, federated, public-sector, critical-infrastructure, sovereign by construction. Counter-UAS airspace awareness is an active EU procurement and grant area (airport disruptions, post-2022 border concern).

A **cheap distributed Remote-ID sensor mesh** is a concrete, demonstrable wedge: low bill of materials, clear public-interest framing, and a fundable open substrate (EU civic/defence programmes, Sovereign Tech Fund resubmission once it runs in production, NLnet for the open mesh layer).

9. Non-technical explainers

For briefings to non-technical audiences (procurement, civic, public-sector), two plain-language pictures:

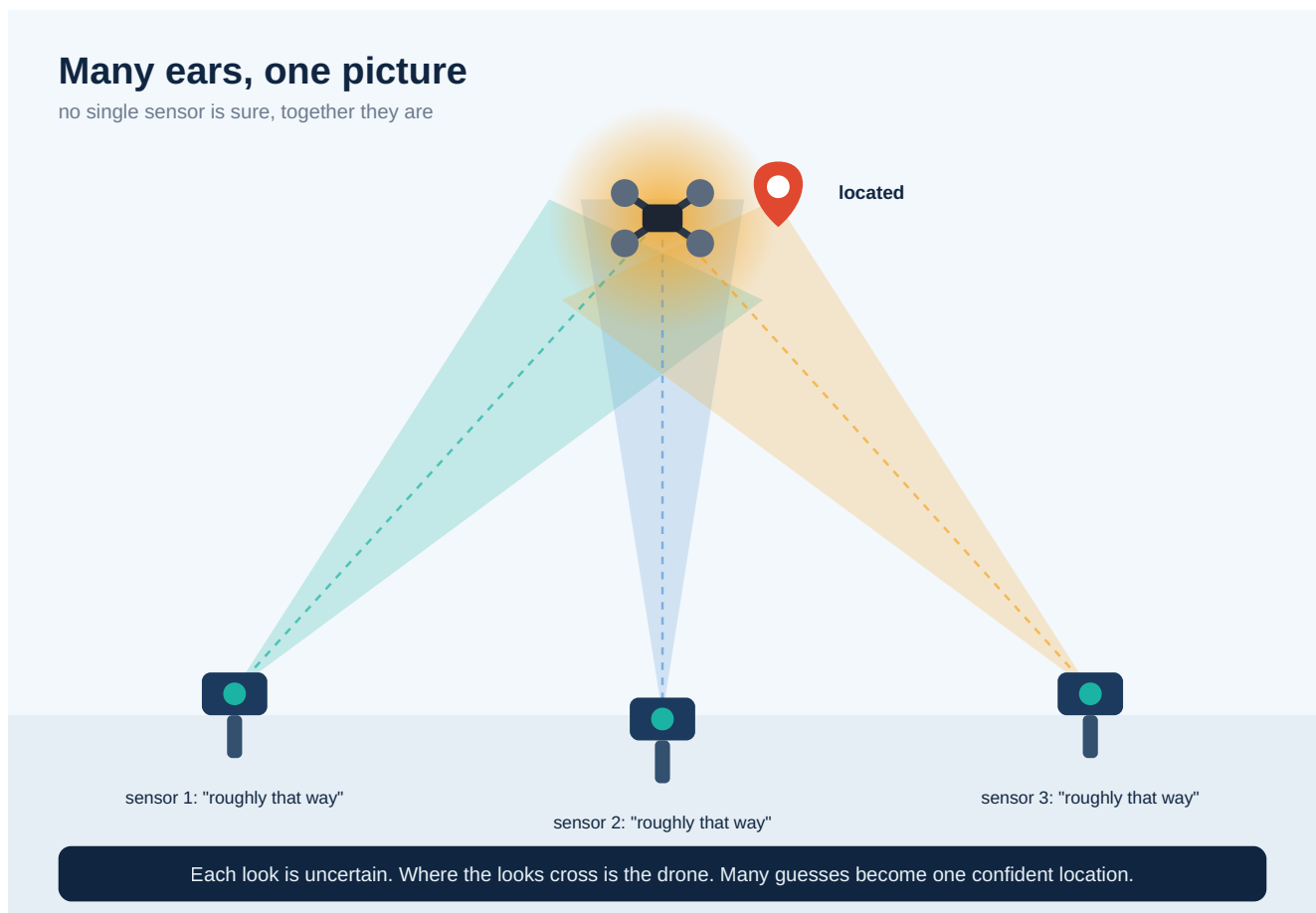
9.1 Neighbourhood watch for the sky



Neighbourhood watch for the sky

Many small, cheap watchers around a site each notice a little. They quietly tell each other what they see. One guard gets a single clear picture. No expensive central tower; no cloud; nothing stops working when the internet drops.

9.2 Many ears, one picture



Many ears, one picture

No single sensor is certain where a drone is. But when several point roughly toward the same spot, the lines cross, and the crossing point is the drone. Many uncertain observations become one confident location.

10. Next steps

1. **Walking skeleton:** `detect_remote_id` division: capability advertisement + `airspace.contact_observed` fact schema + a fusion process manager that correlates two receivers into a single track.
2. **Hardware probe:** ESP32 Remote-ID receiver feeding a Hecate daemon.
3. **One-pager** for the procurement / grant angle, derived from §8.

Appendix A: Fact schema sketch (`airspace.contact_observed`)

```
{
  "fact":      "airspace.contact_observed",
  "v":        1,
  "sensor_id": "remote_id_sensor:perimeter-ne-03",
  "modality":  "remote_id",
  "observed_at": "2026-06-09T15:42:01.214Z",
  "bearing_deg": 47.5,           // optional, sensor dependent
  "range_m":    320,             // optional
  "position":   {"lat": ..., "lng": ...}, // if directly known (Remote ID)
  "drone":      {"id": "...", "type": "DJI Mavic 3", "operator_hint": "..."},
  "confidence": 0.82
}
```

Internal domain events (`contact_detected_v1` etc.) may carry more, and may change shape independently. The fact above is the stable public contract.